

Harshit Mani Tripathi

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Summary

DevOps Intern with production experience, having deployed a live EKS based coding contest platform at Unstop and maintaining a personal hybrid Kubernetes lab focused on cost optimization and zero-trust networking. Skilled in IaC, cloud infrastructure, and end-to-end observability. Recommended by senior engineers, including the Sr. Director and Tech Lead, at Unstop for technical ownership and rapid ramp-up.

Experience

DevOps Intern | Unstop, Delhi

March 2026 - Present

- **Live Platform Launch:** Engineered an EKS-based coding workspace platform, successfully deploying on the production environment for public users.
- **Workspace Orchestration:** Built a FastAPI microservice enabling isolated, persistent coding workspaces across 3 access tiers (organizer/student/evaluator), eliminating per-tenant EC2 provisioning.
- **Cost Optimization:** Reduced coding workspace EBS costs by ~80% via snapshot-based images; identified and proposed ~30-40% additional savings across AWS environment through compute right-sizing and resource cleanup.
- **IaC & Networking:** Provisioned variable-driven EKS clusters via Terraform with Karpenter for dynamic node autoscaling, securing public-facing infrastructure with Traefik.
- **Observability & Scaling:** Built custom SigNoz log preprocessors and developer-facing telemetry dashboards; authored VPA resource-tuning scripts to optimize cluster costs across staging environment.
- **API Design & Integration:** Owned end-to-end workspace lifecycle microservice implementation, iteratively aligning request/response contracts with the backend team.

Infrastructure Projects

Hybrid Infrastructure Lab

github.com/hudater/homelab/

- **System Design:** Designed and implemented a cost-optimized hybrid infrastructure spanning on-premise Kubernetes and cloud-based edge workloads across OCI.
- **Security & Networking:** Enforced zero-trust network isolation and secure routing utilizing OPNsense, NetBird, and VLAN segmentation.
- **Infrastructure and Configuration Automation:** Automated full lifecycle deployment via OpenTofu, managing cloud infrastructure (OCI, k3s, Cloudflare domains and workers) and service configuration-as-code.

Recursive DNS & Observability Stack

blog.hudater.dev/posts/dns-infra/

- **DNS Infrastructure:** Transitioning core recursive DNS routing to Technitium and Unbound, handling ~500k daily queries with a 75% cache hit rate.
- **Centralized Telemetry:** Deployed Prometheus, Grafana, Loki, and Tempo for unified metrics, logs, and distributed tracing across hybrid workloads.

Open Source, Community & Publications

- **Linkme Generator** | github.com/ironicbadger/linkme/pulls?q=author:hudater
Contributed 5+ features in Go and automated multi-stage CI/CD deployments via GitHub Actions.
- **CNCF SRE Games** | showcase.hudater.dev/sre_games.pdf
Investigated and resolved simulated Kubernetes microservice outages under strict time constraints using distributed tracing.
- **Technical Writing** | blog.hudater.dev
Authored in-depth deep-dives covering DNS internal mechanics, core networking protocols, and Git object internals.

Skills

- **Cloud & Infrastructure:** AWS, Oracle Cloud Infrastructure (OCI), Proxmox, Linux
- **Containers & Orchestration:** Kubernetes, Docker, Helm, Karpenter
- **IaC, CI/CD & Version Control:** Terraform (OpenTofu), Ansible, Git, GitHub Actions
- **Languages:** Go, Python, FastAPI, Bash
- **Networking & Security:** OPNsense, NetBird, WireGuard, VPC, VLAN, Traefik, Keycloak, Authentik
- **Observability:** SigNoz, Prometheus, Grafana, Loki, Tempo, Goldilocks (VPA)

Education

B. Tech CSE (Data Science) | Noida International University
8/10 GPA

Aug 2024 - Aug 2028